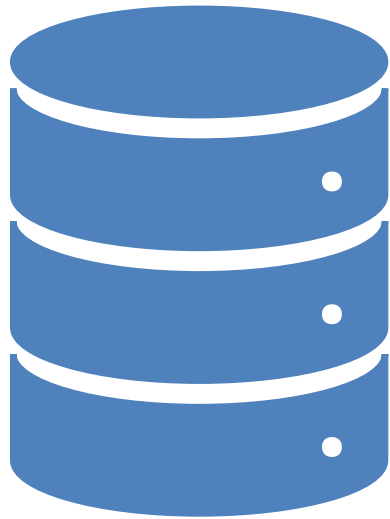




SQL Stored Procedures

A Comprehensive Guide for
Database Developers

[Raghavendra/Data Engineer]

What are Stored Procedures?

Definition:

Precompiled sets of SQL statements saved in the database

Purpose:

Execute multiple SQL statements together as a logical unit

Benefits:

- Improved performance
- Reduced network traffic
- Enhanced security
- Code reusability
- Easier maintenance

Types of Stored Procedures

User-defined Stored Procedures

System Stored Procedures

Temporary Stored Procedures

- Local (#)

- Global (##)

Syntax: Creating a Basic Stored Procedure

```
CREATE PROCEDURE ProcedureName
```

```
AS
```

```
BEGIN
```

```
-- SQL statements go here
```

```
END
```

```
Example:
```

```
CREATE PROCEDURE GetEmployeeData
```

```
AS
```

```
BEGIN
```

```
SELECT * FROM Employees;
```

```
END;
```

```
-- Execute the procedure
```

```
EXEC GetEmployeeData;
```

Real-World Scenario #1: Employee Management



Scenario: HR department needs to add new employees to the system



Stored Procedure: InsertEmployee



- Accepts FirstName, LastName, Department as parameters



- Inserts data into Employees table



- Executes with EXEC InsertEmployee
@FirstName='John', @LastName='Doe',
@Department='IT'



Real- World Scenario: Payroll Processing

Scenario: Finance
team needs to
update employee
salaries

Stored Procedure:
Example available
in documentation

**Real-World
Scenario:
Employee
Offboarding**

Scenario: When employees leave, their records need to be removed

Stored Procedure:
Example available in
documentation



Real-World Scenario: Department Reports

Scenario: Manager
needs reports on
employees by
department

Stored Procedure:
Example available in
documentation

**Real-
World
Scenario:
Sales
Analytics**

Scenario: Sales team needs to analyze product performance

Stored Procedure:
Example available in documentation



Real-World Scenario: Employee Lookup

Scenario: Support team needs to quickly lookup employee details

Stored Procedure:
Example available in
documentation

**Real-World
Scenario:
Department
Transfers**

**Scenario: Updating
employee departments
during reorganization**

**Stored Procedure:
Example available in
documentation**

**Real-
World
Scenario:
Workforce
Analytics**

**Scenario: Company
dashboard needs
employee headcount**

**Stored Procedure:
Example available in
documentation**

Real-World
Scenario:
Inventory
Management

Scenario: Adding
new products to the
inventory system

Stored Procedure:
Example available in
documentation



Real-World Scenario: Premium Product Analysis

Scenario: Marketing team needs to identify high-value products

Stored Procedure:
Example available in documentation

When to Use Stored Procedures?

Complex Business Logic: When you need to execute multiple SQL statements

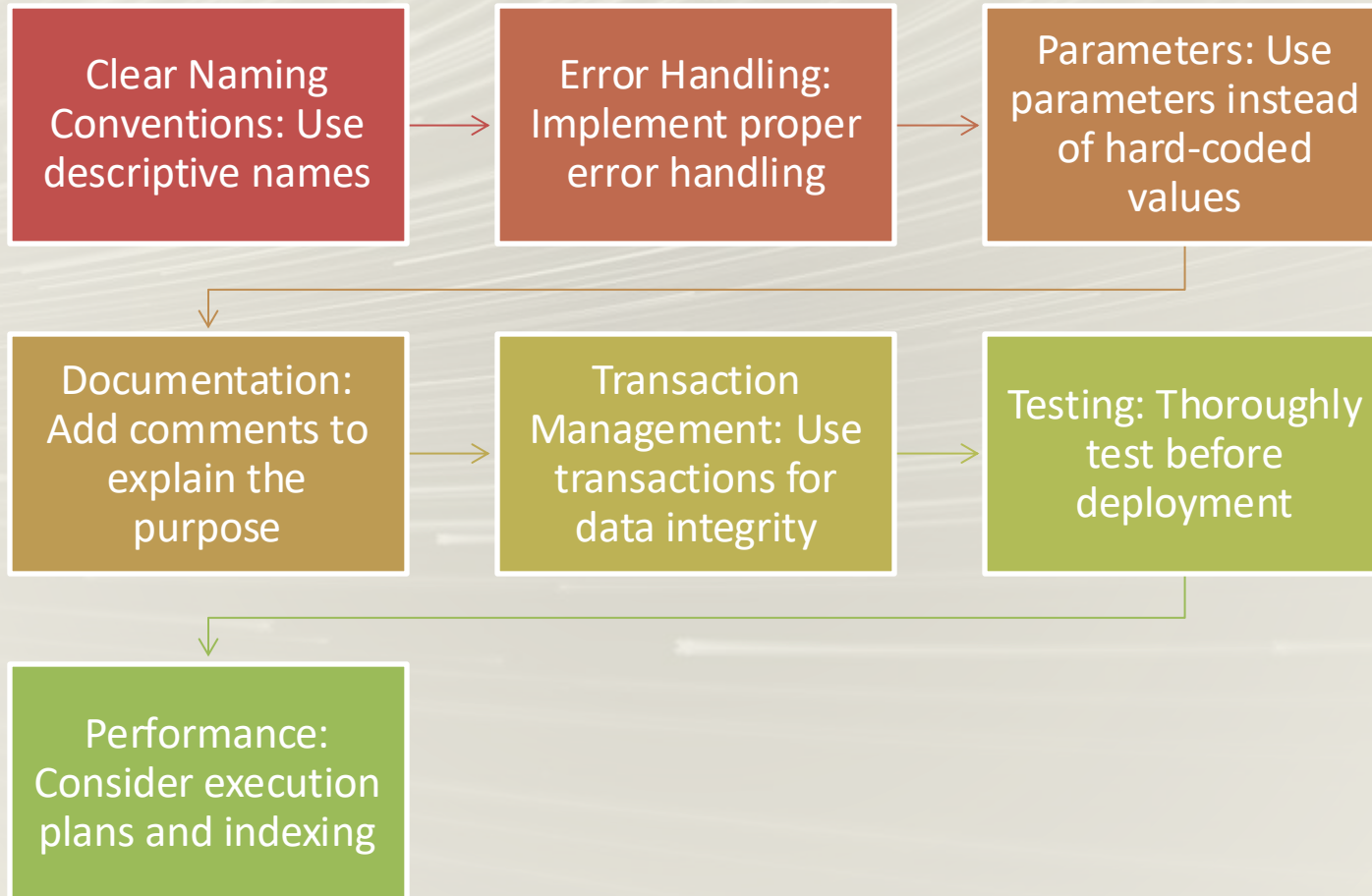
Data Validation: Ensuring data integrity before insertion

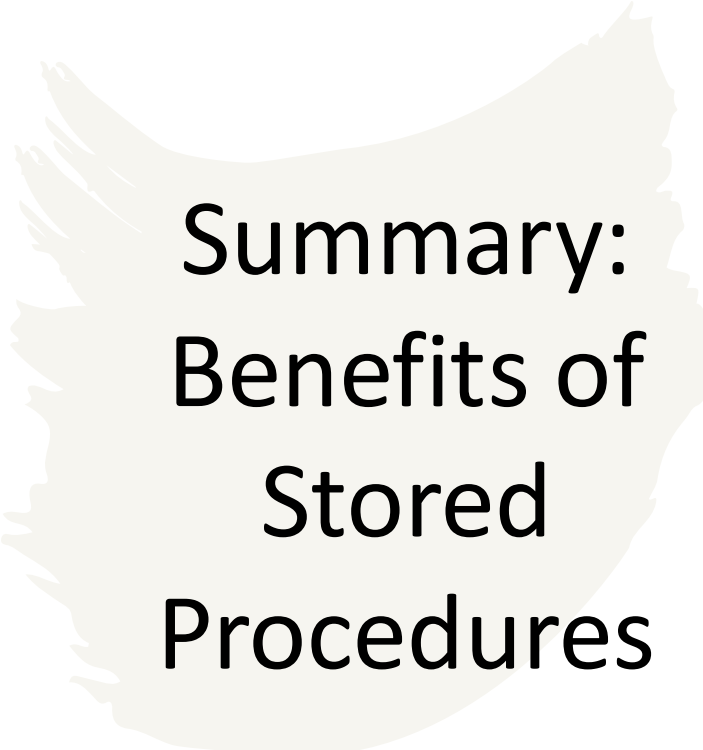
Performance Critical: For frequently executed queries

Security: Control access to sensitive data

Maintenance: Easier to update logic in one place

Best Practices for Stored Procedures





Summary: Benefits of Stored Procedures

Performance Optimization: Reduced network traffic and precompiled execution

Code Reusability: Write once, use many times

Security Enhancement: Grant execute permissions without table access

Maintainability: Centralized business logic

Parameterization: Adaptable to different scenarios

Reduced Network Traffic: Only procedure name and parameters sent



Thank You!

Connect with me for more SQL and
database content!

[LinkedIn Profile and Contact Information]
[https://www.linkedin.com/in/raghavendra-
marakattu-b0300113b/](https://www.linkedin.com/in/raghavendra-marakattu-b0300113b/)

#SQLTips #DatabaseDevelopment
#StoredProcedures #DataEngineering